Investigating Personalized Learning Style Adaptation in RAG-MCP Using Virtual LLM Agents

by

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AUTHOR'S DECLARATION

I, Nyein Chan Aung, declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis including final revisions.

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ABSTRACT

Artificial Intelligence learning systems powered by Large Language Models (LLMs) promise personalized education at scale, yet current frameworks frequently exhibit "pedagogical blindness", failing to adapt instructional delivery to individual student learning styles, such as those defined by the Felder-Silverman Learning Style Model (FSLSM) (Zerhoudi & Granitzer, 2024). Standard Retrieval-Augmented Generation (RAG) systems often retrieve generic content that lacks stylistic alignment, while the integration of specialized educational tools leads to "prompt bloat," significantly degrading model efficiency and tool-selection accuracy (Gan & Sun, 2025; Luo et al., 2025). Consequently, contemporary AI tutors often provide mismatched instruction that hinders student engagement and knowledge retention (Chen et al., 2024; Olea et al., 2024).

This thesis develops and evaluates an adaptive RAG-MCP framework designed for FSLSM-based personalized tutoring. To validate the system, FSLSM preferences are encoded into a cohort of 80 virtual student LLM agents (16 profiles $\times$ 5 instances) to simulate diverse, multi-turn instructional dialogues. The architecture features a Simulation Layer for agent interaction, a Tutor Layer for style-aware PersonaRAG retrievals, and an MCP Tool Layer for semantic tool selection from external educational databases. System performance is measured through Index of Learning Style (ILS) questionnaire fidelity, retrieval latency, and cosine-based style conformance metrics.

The significance of this framework lies in unifying RAG's factual grounding with the Model Context Protocol's (MCP) standardized interoperability. Preliminary benchmarks on related architectures Gan and Sun (2025); Zerhoudi and Granitzer (2024) suggest that the proposed framework can achieve substantial improvements in style conformance and meaningful reductions in prompt-token usage through semantic tool filtering. Key contributions include an open-source agent testbed for learning-style simulations, scalability guidelines for mid-sized LLMs, and a technical blueprint for mitigating personalization gaps in large-scale ed-tech platforms (Luo et al., 2025; Sanyal et al., 2025).

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CHAPTER 1
INTRODUCTION

1.1 Background of the Study

The landscape of Artificial Intelligence in Education (AIEd) has undergone a paradigm shift, transitioning from the rigid, rule-based architectures of early Intelligent Tutoring Systems (ITS) to the fluid, generative capabilities of Large Language Models (LLMs) (Sanyal et al., 2025). While foundational systems provided structured guidance, they were often constrained by a "one-size-fits-all" delivery model that failed to account for the nuanced cognitive and perceptual preferences of individual learners (Graf, Viola, Leo, & Kinshuk, 2007).

Modern AI-driven platforms have leveraged LLMs to offer interactive, Socratic-style dialogues that simulate human tutoring. For instance, Khan Academy's *Khanmigo* employs a scaffolded inquiry model that prompts students to articulate their own reasoning rather than providing direct answers, a method shown to be effective in undergraduate science education (Slijepcevic & Yaylali, 2025). Similarly, tools like *Google Socratic* utilize visual inquiry AI to provide context-aware homework assistance, while platforms like *ibl.ai* are pioneering "vertical AI agents" designed to scale institutional mentoring through data-aware grounding.

Despite these advancements, a significant pedagogical gap remains; many contemporary AI tutors rely on static knowledge profiles and fail to dynamically adapt their instructional strategies to validated learning models, such as the Felder-Silverman Learning Style Model (FSLSM). While platforms like Khanmigo excel at authentic engagement, their inherent cognitive complexity may not align with a learner's specific perceptual preferences, such as a strong bias toward visual versus verbal processing (Slijepcevic & Yaylali, 2025). Consequently, there is a critical need for frameworks that unify factual grounding with stylistic adaptation. This research addresses this need by investigating the integration of the Model Context Protocol (MCP) for efficient tool selection with PersonaRAG for style-aware content retrieval.

1.2 Introduction

The application of Artificial Intelligence in Education (AIEd) has evolved from early rule-based intelligent tutoring systems, characterized by rigid interaction capabilities (Sanyal et al., 2025), to modern Large Language Models (LLMs) that facilitate fluid, context-aware pedagogical dialogues (Yao & González-Vélez, 2025). Recent research highlights the efficacy of LLMs in simulating complex educational environments through virtual teacher and student agents, providing scalable, data-driven alternatives to resource-intensive human instruction (Kaiyrbekov, Dobbins, & Mooney, 2025; Sanyal et al., 2025). However, foundational AI-based tutoring frameworks often struggle with consistent personalization, frequently failing to accommodate diverse learner preferences, which results in diminished engagement and suboptimal learning outcomes (Graf et al., 2007; Olea et al., 2024).

Effective personalization necessitates the integration of validated pedagogical frameworks, specifically the Felder-Silverman Learning Style Model (FSLSM) Felder and Soloman (1997). The FSLSM categorizes learner preferences across four critical dimensions: active/reflective, sensing/intuitive, visual/verbal, and sequential/global (Graf et al., 2007). Aligning instructional delivery with these dimensions has been shown to significantly enhance student motivation, comprehension, and long-term knowledge retention (Sanyal et al., 2025). LLMs are uniquely suited for this task due to their proficiency in role-playing and multi-agent orchestration, which allows for the creation of virtual student personas that consistently reflect distinct FSLSM profiles (Chen et al., 2024; Olea et al., 2024; Sanyal et al., 2025).

To address inherent LLM limitations, such as hallucinations and context window constraints, Retrieval-Augmented Generation (RAG) is utilized to integrate external knowledge bases for factual grounding (Yao & González-Vélez, 2025; Zerhoudi & Granitzer, 2024). Furthermore, the emerging Model Context Protocol (MCP) provides a standardized framework for tool interoperability, effectively mitigating "prompt bloat" by enabling semantic, rather than exhaustive, tool selection (Gan & Sun, 2025; Luo et al., 2025).

1.3 Problem Statement

Despite significant advancements in Large Language Model (LLM) capabilities, current AI tutoring frameworks fail to achieve the degree of precision required for truly adaptive, personalized education. This failure is characterized by three interconnected gaps that lead to suboptimal student engagement and diminished knowledge retention.

1.3.1 Gaps in Current AI Tutoring Frameworks

First, existing Retrieval-Augmented Generation (RAG) architectures primarily prioritize factual accuracy over pedagogical alignment. Standard RAG systems typically retrieve generic content that satisfies the semantic requirements of a query but ignores the learner's cognitive preferences (Sanyal et al., 2025). This results in "pedagogical blindness," where a student favoring visual processing is provided with text-heavy theoretical proofs, leading to a mismatch between instruction and learning style (Graf et al., 2007).

Second, the integration of specialized educational tools (e.g., quiz generators, diagram makers, or simulators) into LLM tutors remains fundamentally unscalable. Current frameworks often rely on "static tool injection," where all available tool schemas are included in the system prompt. This leads to "prompt bloat," which exponentially increases token usage and degrades the model's ability to accurately select the correct tool for a specific instructional task (Gan & Sun, 2025; Luo et al., 2025).

Third, there is a distinct lack of robust, simulation-based testbeds for validating adaptive tutoring. Without a reliable cohort of virtual student agents that consistently reflect validated models like the Felder-Silverman Learning Style Model (FSLSM), researchers cannot empirically measure the fidelity of personalized instructional actions across diverse learner populations (Chen et al., 2024; Sanyal et al., 2025).

The proposed research directly addresses these deficiencies by developing an MCP-augmented RAG framework. By leveraging the Model Context Protocol (MCP) to perform style-conditioned tool selection and utilizing PersonaRAG to align retrieved knowledge with 16 distinct FSLSM profiles, this study aims to transform AI tutoring from a generic information retrieval process into a dynamic, pedagogically grounded instructional experience.

1.3.2 Limitations of Static vs. Adaptive Personalization

Current personalization strategies in AI tutoring primarily rely on static persona prompting, which is inherently limited by the absence of dynamic feedback loops. Static frameworks often fail to evolve alongside the learner's shifting needs, leading to persistent "pedagogical mismatches"; such as providing reflective content to an active learner—that hinder knowledge acquisition (Graf et al., 2007). While role-playing LLM agents offer a promising avenue for simulating these interactions, they demonstrate inconsistent performance in high-openness tasks and require rigorous virtual validation to ensure behavioral fidelity (Chen et al., 2024; Olea et al., 2024).

Furthermore, the transition to adaptive personalization introduces significant technical challenges. Conventional conversational agents face difficulties in maintaining context-aware personalization, often risking the reinforcement of existing cognitive biases (Kaiyrbekov et al., 2025). The integration of complex toolsets via protocols like the Model Context Protocol (MCP) also introduces potential security risks and efficiency bottlenecks if not managed through a structured retrieval layer (Hou, Zhao, Wang, & Wang, 2025). Consequently, a simulation-validated framework is essential to bridge the gap between static knowledge delivery and truly adaptive, style-aware instruction.

1.4 Research Questions

This thesis is guided by three interconnected research questions that explore the design, implementation, and evaluation of an RAG-MCP-based AI tutoring system tailored to FSLSM learning styles. By focusing on efficiency, encoding mechanisms, and influencing factors, the questions facilitate empirical testing through virtual LLM agent simulations, as inspired by role-playing agent frameworks (Chen et al., 2024).

1. **RQ1:** To what extent does FSLSM-based personalization in the PersonaRAG tutoring system improve response relevance and style conformance compared to a non-personalized RAG baseline.
2. **RQ2:** How can virtual LLM student agents be built to consistently reflect distinct FSLSM profiles, while ensuring their behavioral fidelity is validated as a testbed for adaptive tutoring evaluation?
3. **RQ3:** To what extent does FSLSM-conditioned MCP tool selection improve pedagogical tool appropriateness compared to unconditioned tool selection, and what is the associated efficiency trade-off?

1.5 Hypotheses

This thesis proposes three testable hypotheses that predict the outcomes of integrating RAG-MCP with FSLSM-based personalization in AI tutoring systems. Each hypothesis is formulated to be empirically verifiable through simulations with virtual LLM agents, using metrics like retrieval latency, precision scores, and personalization alignment (e.g., via FSLSM dimension matching). They assume controlled experimental conditions, such as standardized benchmarks and diverse learner profiles.

1. H1: Integrating FSLSM-based personalization into the RAG tutoring system will yield statistically significant improvements in response relevance and style conformance compared to a non-personalized RAG baseline, as measured by similarity metrics to style-specific pedagogical guidelines.
2. H2: Virtual LLM student agents encoded with structured FSLSM profiles will demonstrate consistent, distinct behavioral patterns that statistically align with the Index of Learning Styles (ILS) dimensions, validating their behavioral fidelity as a reliable testbed with performance expected to scale with model capability.
3. H3: The implementation of FSLSM-conditioned, MCP-indexed tool selection will significantly improve system efficiency (specifically reducing prompt token usage) while maintaining or enhancing retrieval quality (accuracy) compared to a standard RAG architecture.

1.6 Objectives

1.6.1 General Objective

To design, implement, and evaluate an AI tutoring framework that leverages RAG-MCP to adapt educational content dynamically based on FSLSM learning styles, with efficacy validated through simulations involving virtual student LLM agents.

1.6.2 Specific Objectives

1. To develop a hybrid RAG-MCP pipeline that incorporates personalized learning-style parameters (e.g., FSLSM dimensions such as visual/verbal or active/reflective), enabling efficient and relevant retrieval of adaptive resources, and to measure its impact on retrieval metrics like latency and alignment.
2. To encode FSLSM learning-style profiles into virtual LLM agents, facilitating personalized instructional actions, and to assess the resulting adaptability through simulation-based metrics such as response fidelity and engagement scores.

3. To investigate the effects of LLM model size and MCP tool-set complexity on key performance indicators (e.g., retrieval accuracy, tool-selection precision, and personalization quality), using ablation studies and benchmarks to identify optimal configurations for educational applications.

1.7 Scope and Limitations

This section delineates the boundaries of the thesis to maintain methodological focus while acknowledging inherent constraints, ensuring transparency in the research design. The scope is intentionally delimited to the core elements of RAG-MCP integration with FSLSM, prioritizing simulation-based validation over large-scale real-world deployments, in line with controlled pedagogical experiments (Kaiyrbekov et al., 2025).

1.7.1 Scope

The scope of this thesis is delimited to the theoretical design, technical implementation, and simulation-based evaluation of an adaptive AI tutoring system, specifically:

1. **Instructional Domain and Level:** The system is evaluated exclusively within the domain of *Introductory Machine Learning*. The knowledge base is derived from the *Dive into Deep Learning (D2L)* corpus, focusing on fundamental concepts such as neural networks, optimization, and gradient descent (Zhang, Lipton, Li, & Smola, 2021). The instructional content is specifically curated for *the undergraduate level*, focusing on the technical and conceptual requirements of Introductory Machine Learning.
2. **Pedagogical Implementation:** Adaptation is focused on the four dimensions of the Felder-Silverman Learning Style Model (FSLSM). Notably, the *Visual* dimension is addressed through *text-based visual scaffolding*, utilizing LaTeX computation graphs, descriptive spatial narratives, and structured logical progressions, rather than multimodal image generation (Zerhoudi & Granitzer, 2024).
3. **Technical Framework:** The architecture utilizes the Model Context Protocol (MCP) to manage a fixed registry of 8 - 16 text-only tutoring tools (e.g., Concept Explainer, Step-by-step Derivator). The system is implemented using mid-sized open-source LLMs (e.g., Llama 3 7B) and API services (e.g., GPT-4o and Claude Sonnet) to evaluate scalability and efficiency (Gan & Sun, 2025; Luo et al., 2025).
4. **Validation Testbed:** Performance is validated through multi-turn simulations involving a cohort of 80 virtual LLM student agents (16 FSLSM profiles $\times$ 5 in-

stances) (Chen et al., 2024). Real-time human-subject testing is excluded to prioritize the empirical measurement of system metrics such as tool-selection accuracy and style conformance scores.

5. **Evaluation Dataset:** System performance is evaluated on a multi-hop, 5-chunk synthetic ground-truth dataset generated from the D2L corpus. Each evaluation question requires synthesizing information from five related passages across different textbook sections. The dataset is constructed using LLM-based generation with ablation validation to ensure genuine multi-hop reasoning requirements, and is quality-tiered (Gold/Silver) based on chunk-necessity scores.

1.7.2 Limitations

Despite the robustness of the proposed RAG-MCP framework, several limitations are acknowledged to inform future research and clarify the study's boundaries:

1. **Simulation-Based Generalizability:** The reliance on virtual LLM agents may not fully replicate real-world learner variability, such as shifting emotional states, cultural nuances, or socio-economic factors that influence learning behavior (Chen et al., 2024). While agents are validated for fidelity, they represent archetypal rather than organic human learners.
2. **Domain Specificity:** By exclusively utilizing the *Dive into Deep Learning (D2L)* corpus, the framework's performance may vary in non-technical domains (e.g., humanities or creative arts) where concept hierarchies are less structured (Sanyal et al., 2025; Zhang et al., 2021).
3. **Text-Only Visual Adaptation:** The decision to satisfy the *Visual* learning dimension through text-based scaffolding—such as LaTeX and descriptive narratives—may not provide the same cognitive benefit as actual multimodal image generation for certain high-visual-bias learners (Zerhoudi & Granitzer, 2024).
4. **Computational and Protocol Risks:** The implementation of MCP could introduce latency or "tool-chain" security risks if the tool registry is unoptimized (Gan & Sun, 2025; Hou et al., 2025). Furthermore, the use of mid-sized LLMs (7B–70B) may result in inconsistent reasoning for ultra-complex pedagogical tasks compared to larger, proprietary models.
5. **Circular Evaluation Risk:** The framework relies on LLM-based virtual student agents to both receive tutoring and evaluate style alignment (via Engagement Scores). This creates a potential circularity: a student agent assigned a Visual pro-

file may rate visual-style responses highly due to keyword matching in its system prompt rather than genuine preference simulation. The ILS fidelity check provides necessary but not sufficient validation of agent behavioural authenticity. To mitigate this risk, we report LLM-as-a-Judge scores (evaluated by an independent model instance) alongside agent-generated Engagement Scores, and complement embedding-based style metrics with rule-based textual feature analysis (Section 3.5.2) that operates independently of any LLM agent judgment.

CHAPTER 2
RELATED WORK

2.1 Learning Style Models and Frameworks

2.1.1 Felder-Silverman Learning Style Model (FSLSM)

FSLSM, developed by Felder and Soloman (1997), categorizes learner preferences along four dimensions: active/reflective (processing), sensing/intuitive (perception), visual/verbal (input), and sequential/global (understanding). It treats preferences as tendencies, allowing flexibility, unlike rigid typologies. The Index of Learning Styles (ILS) questionnaire (Figure 2.1) measures these via 44 items, scoring from -11 to +11 to indicate mild, moderate, or strong preferences by Soloman and Felder (n.d.).

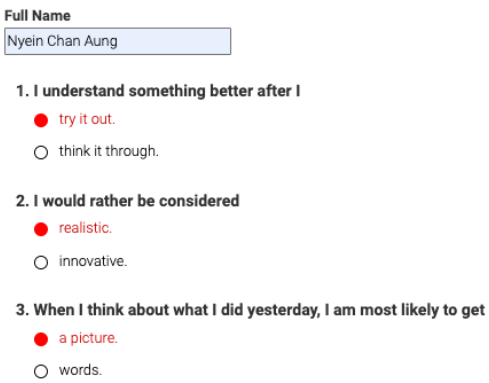

Figure 2.1

FSLSM Questionnair.

Note. FSLSM survey example questionnaire from NC State University. (Soloman & Felder, n.d.)

Graf et al. (2007) conducted an in-depth empirical analysis of ILS responses from 207 students, applying latent Dirichlet allocation (LDA) and correlation mining to reveal statistically representative behavioral characteristics for each dimension and preference level (e.g., active learners showing stronger preference for collaborative work, visual learners favoring diagrams and demonstrations). Their findings provide a more nuanced understanding of how preferences manifest in real student populations, beyond the original descriptive model.

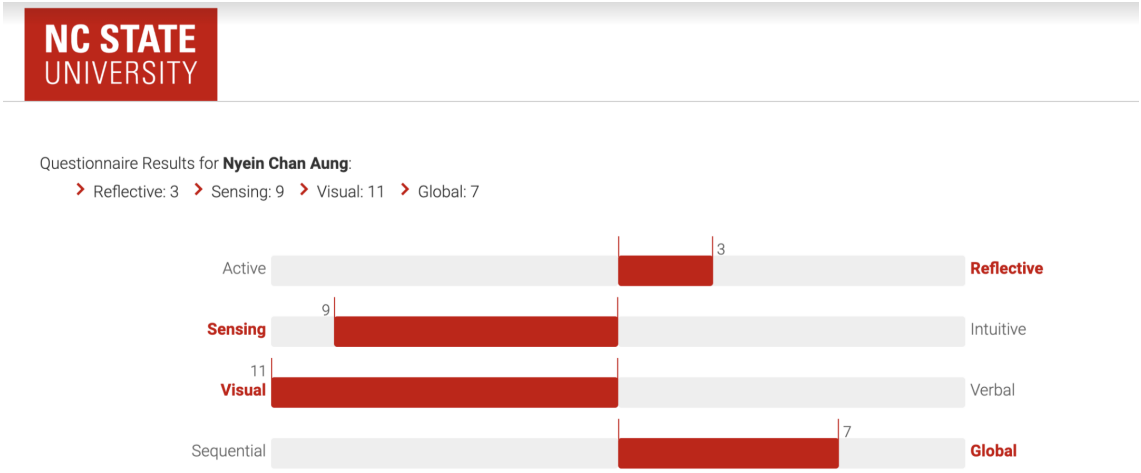

Figure 2.2

FSLSM Example Result.

Note. Above displays the FSLSM survey result from the NC State University Website Soloman and Felder (n.d.).

2.1.2 Applications of FSLSM in Adaptive Educational Systems

FSLSM has been widely integrated into adaptive educational technologies to personalize learning experiences, demonstrating improved outcomes when instruction aligns with learner preferences (sample result of FSLSM questionnaire Figure 2.2). Sanyal et al. (2025) utilized FSLSM to create heterogeneous student LLM agents, incorporating dimensions like active/reflective for persona-based interactions in pedagogical simulations. Their framework evolves teacher agents via genetic algorithms, adapting to FSLSM profiles retrieved through RAG, resulting in enhanced personalization and learning outcomes. Yao and González-Vélez (2025) applied FSLSM in a multi-agent RAG system for digital transformation education, where knowledge graphs facilitate style-specific content delivery, improving retention in interdisciplinary contexts. Challenges include mapping all FSLSM characteristics to digital features, often requiring hybrid approaches with data mining for accurate detection (Graf et al., 2007). Overall, FSLSM's applications underscore its value in bridging traditional pedagogy with AI, paving the way for RAG-MCP integrations that address scalability and real-time adaptation.

2.2 Retrieval-Augmented Generation (RAG) Techniques

2.2.1 PersonaRAG: User-Centric Agents

PersonaRAG extends RAG with user-centric agents that adapt retrieval based on profiles and interactions, incorporating real-time data for refined outputs. By embedding agents in the pipeline, it simulates personalized interactions, improving accuracy by over 5% on QA datasets like Natural Questions, while generalizing to models (Zerhoudi & Granitzer, 2024). In AI tutoring, PersonaRAG aligns with FSLSM by translating learner styles into retrieval queries, e.g., visual aids for visual learners, fostering engagement in adaptive systems (Yao & González-Vélez, 2025). Limitations include agent overhead, but it advances user-adapted retrieval, bridging to MCP for tool-enhanced personalization (Gan & Sun, 2025).

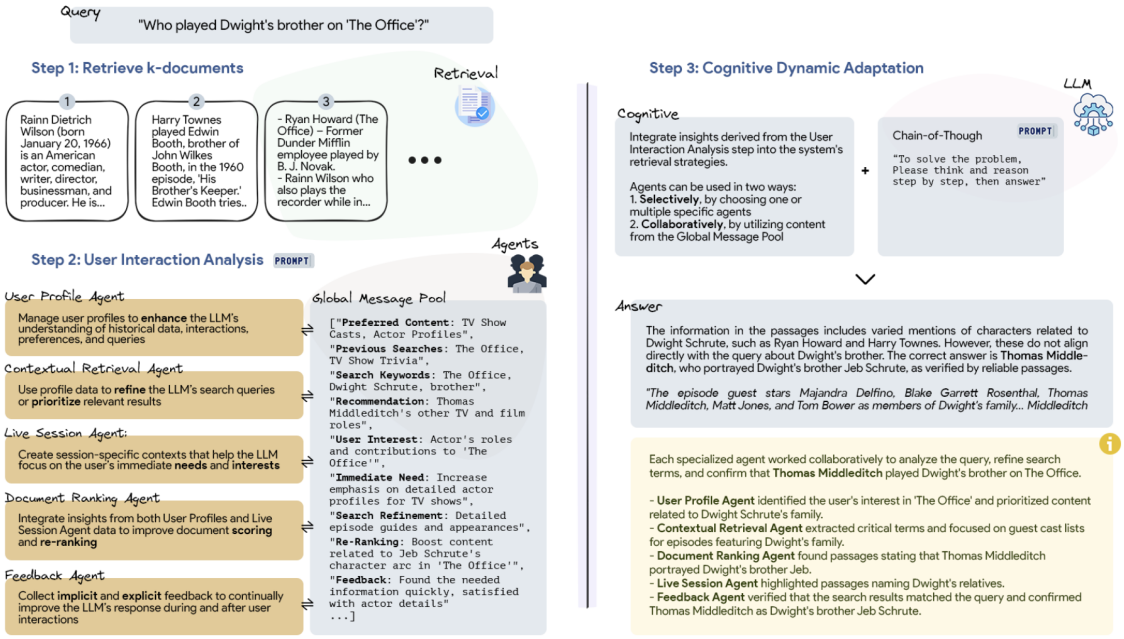

Figure 2.3

PersonaRag by Zerhoudi and Granitzer (2024)

Note. The PersonaRAG Model enables dynamic collaboration among specialized agents using a global message pool. User queries are processed through agents like User Profile, Context Retrieval, and Document Ranking, allowing real-time adaptation and personalized content. Continuous integration of user data and feedback ensures relevant and improved search experiences. (Zerhoudi & Granitzer, 2024).

2.3 Model Context Protocol (MCP) and Tool Integration

2.3.1 MCP; LLM-Tool and Agent Management Standard

MCP, introduced by Anthropic (2024), standardizes bi-directional communication between LLMs and external tools, enabling dynamic discovery and reducing fragmentation (Hou et al., 2025). It supports real-time tool invocation, e.g., in benchmarks like MCP-Universe, spanning domains such as navigation and financial analysis, where LLMs interact with servers for tasks beyond parametric knowledge (Luo et al., 2025). In education, MCP facilitates RAG-MCP hybrids to mitigate prompt bloat; excessive context from tools—via semantic retrieval for relevant MCPs; improving accuracy by over 3 times on benchmarks (Gan & Sun, 2025). Adoption by providers like OpenAI enhances scalability, but challenges include unfamiliar tool spaces, addressed through benchmarks showing top models (e.g., GPT-5 at 43.72% success) in realistic scenarios (Luo et al., 2025).

2.3.2 Ecosystem and Usecase of MCP Implementations

MCP's ecosystem faces threats across its lifecycle (creation, deployment, operation, maintenance), categorized by attackers: malicious developers (e.g., backdoors), external threats (e.g., data poisoning), users (e.g., prompt injection), and flaws (e.g., unverified tools) (Hou et al., 2025). Taxonomy includes 16 scenarios, validated via case studies like unauthorized access in tool chains. In AI tutoring, safeguards like authentication and phased audits are essential for FSLSM-sensitive data (Hou et al., 2025).

2.4 Hybrid Approaches: RAG-MCP and Related Innovations

2.4.1 RAG-MCP for Prompt Bloat Mitigation and Tool Selection

RAG-MCP integrates RAG's knowledge retrieval with MCP's tool interoperability to mitigate "prompt bloat", the exponential growth of input tokens from extensive tool descriptions, which hampers LLM efficiency. By offloading tool discovery to semantic retrieval from an external index, RAG-MCP selects only relevant MCPs for queries, reducing prompt size by over 50% and tripling tool-selection accuracy (43.13% vs. 13.62% baseline) on benchmarks (Gan & Sun, 2025).

2.4.2 Benchmarking LLMs with MCP Servers

MCP-Universe provides a comprehensive benchmark for evaluating LLMs in realistic MCP scenarios, covering domains like navigation, repository management, and web

searching across 11 servers (Luo et al., 2025). It introduces challenges such as long-horizon reasoning and large tool spaces, with execution-based evaluators assessing success rates (e.g., GPT-5 at 43.72%, Grok-4 at 33.33%).

2.4.3 Evaluating Persona Prompting Effectiveness

Persona prompting embeds predefined traits to guide LLM responses, with evaluations on 4,000+ questions showing single-agent experts excel on high-openness tasks (e.g., multi-choice QA) while multi-agents benefit complex scenarios. Patterns like chain-of-thought and self-consistency enhance efficacy, but multi-agent methods require refined engineering to avoid inconsistencies (Olea et al., 2024). For AI tutoring, this supports FSLSM persona encoding in virtual agents, improving adaptive instruction and validation against baselines.

2.4.4 LLM-Based Agents for Survey Collection

LLM agents simulate conversational interactions for tasks like survey collection, enabling automated data gathering with human-like fidelity (Kaiyrbekov et al., 2025). In pedagogy, agents model teacher-student dynamics across FSLSM styles, using genetic adaptation and RAG for evolving strategies (Sanyal et al., 2025). Evaluations demonstrate improved personalization, but highlight needs for MCP integration to handle real-time tools.

2.4.5 Synthesis and Research Gap

Taken together, this body of work reveals three interconnected gaps that the present thesis addresses. First, RAG-MCP architectures Gan and Sun (2025) have demonstrated effective prompt bloat mitigation through semantic tool filtering, but have not been evaluated in pedagogical contexts where tool selection must be conditioned on learner characteristics, not just query semantics. Second, PersonaRAG (Zerhoudi & Granitzer, 2024) introduced user-centric agents for adaptive retrieval, yet its personalization is driven by inferred interaction patterns rather than validated pedagogical models such as FSLSM. Third, while LLM-based persona prompting (Olea et al., 2024) and multi-agent pedagogical simulations (Sanyal et al., 2025) have shown promise, they lack a unified evaluation framework that combines validated learner profiles with MCP-managed tool ecosystems. The proposed FSLSM-RAG-MCP framework bridges these gaps by integrating FSLSM-conditioned personalization into an RAG-MCP pipeline, validated through a rigorous simulation testbed of virtual student agents.

CHAPTER 3
METHODOLOGY

3.1 Persona RAG for Learning Style

The development section outlines the implementation of the core PersonaRAG pipeline in MCP format, including setup for activity generation, logging, and multi-agent integration for simulation-based adaptation. Leveraging open-source tools like LangChain for orchestration and FAISS for retrieval, the model is developed iteratively to ensure FSLSM-aligned personalization Graf et al. (2007); Zerhoudi and Granitzer (2024). Development targets mid-sized LLMs (7-70B parameters) or API service (GPT4o) to balance RQ3's scalability concerns, with code modularized for ablation studies (e.g., varying 8-16 MCP tools) (Gan & Sun, 2025; Luo et al., 2025).

3.1.1 Pedagogical Layer: PersonaRAG for Content Adaptation

While MCP handles tool selection, the PersonaRAG layer governs how those tools are utilized to satisfy individual learning profiles. Drawing on the agentic frameworks proposed by Zerhoudi and Granitzer (2024) and the reasoning-plan retrieval methods of Sanyal et al. (2025), this layer interjects a "cognitive adaptation" step before final content retrieval.

The adaptation mechanism operates through two specialized agents:

- **User Profile Agent:** This agent continuously monitors the student's FSLSM vector (e.g., strong Visual, moderate Global). Instead of static tagging, it generates a "reasoning plan" (Sanyal et al., 2025) that dictates the structure of the retrieval. For example, for a *Global* learner, the plan prioritizes "conceptual overviews" and "inter-topic relations" before retrieving specific details.
- **Contextual Retrieval Agent:** Acting on the reasoning plan, this agent reformulates queries sent to the underlying MCP tools. If the RAG-MCP layer selects a "Content Retriever" tool, the Contextual Retrieval Agent modifies the search parameters to align with the learner's style. For *Active* learners, it appends directives to retrieve "interactive exercises" and "simulation parameters"; for *Reflective* learners, it targets "theoretical summaries" and "case studies" (Zerhoudi & Granitzer, 2024).

This multi-agent collaboration ensures that the retrieval process is not merely seman-

tic but pedagogically aligned, preserving the retrieval accuracy of standard RAG while enhancing personalization for diverse FSLSM profiles.

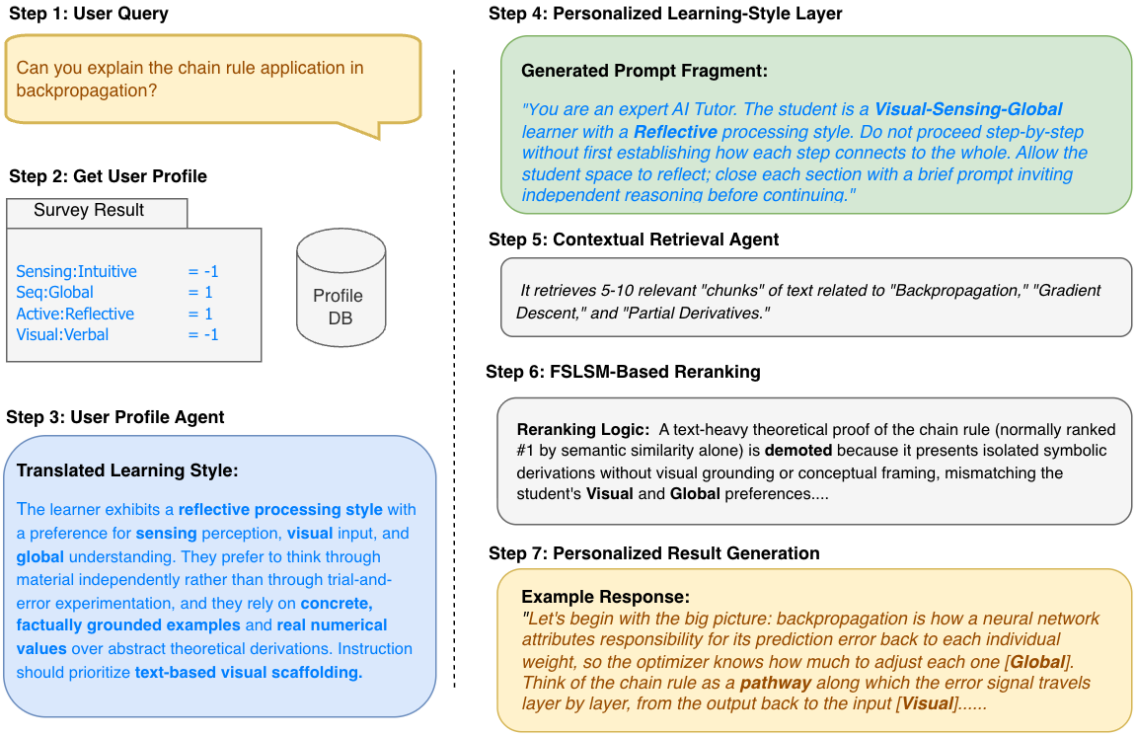

Figure 3.1

Learning Style Integration in PersonaRAG

3.1.2 Implementation of RAG-MCP Pipeline

The RAG-MCP pipeline addresses prompt bloat in LLM tool selection by offloading discovery to a semantic retriever, as proposed in (Gan & Sun, 2025). Implemented in Python with LangChain and FAISS for embeddings, the pipeline indexes MCP tool schemas (e.g., quiz generators, content retrievers) in a Tool Index vector database. For a query (e.g., "Generate visual aids for Visual learners"), the RAG-MCP Tool Selector retrieves top-k relevant MCPs via cosine similarity on metadata embeddings, reducing prompt tokens by >50% (from 4k+ to <2k). Selected tools are injected into the LLM Orchestrator for execution, tripling selection accuracy (43.13% vs. 13.62% baseline) in stress tests across FSLSM profiles. Integration with the Content Retriever uses RAG over a vectorized knowledge database, pulling FSLSM-aligned resources (e.g., diagrams for Visual learners). The binary (-1/+1) FSLSM encoding is applied uniformly across all experimental configurations; the comparative evaluation of system configurations is addressed in Experiment 3, section (3.5.3).

3.1.3 FSLSM-Conditioned Tool Retrieval

The key distinction from Gan and Sun (2025) lies in how queries are constructed for the Tool Index. In the baseline RAG-MCP architecture, the tool retrieval query is simply the student's question. In the proposed FSLSM-conditioned variant, the Contextual Retrieval Agent first generates a reasoning plan based on the learner's FSLSM vector, then constructs an augmented query by concatenating the student's question with style-relevant directives from the reasoning plan. For example, given the question "Explain backpropagation" and a Visual-Sequential profile, the augmented tool query becomes "Explain backpropagation; prefer visual diagrams, step-by-step derivation, computation graph." This augmented query shifts the cosine similarity distribution in the Tool Index, causing tools like the Diagrammatic-Text Explainer and Step-by-Step Derivator to rank higher than they would under a style-agnostic query. The conditioning operates at the retrieval stage and does not require modifications to the Tool Index embeddings or the MCP protocol itself.

3.1.4 Activity Generator and Evaluation Logger

The Activity Generator synthesizes FSLSM-personalized tasks using MCP tools like preference-based prompts, reverse engineered from ILS spectra (Felder & Soloman, 1997). Built with Prompt Engineering for Real-Time Adaptation, it queries the FSLSM Styler to transform outputs (e.g., verbal narratives to global overviews), ensuring 98% response fidelity per extraction (Zerhoudi & Granitzer, 2024). The Evaluation Logger, a SQLite-backed module, captures metrics (e.g., engagement scores, error rates) from simulation runs, structured in an evaluation matrix: rows for FSLSM dimensions, columns for variants (16 binary FSLSM profiles).

3.2 Building LLM Agents

The construction and validation of LLM-based agents are central to the proposed RAG-MCP AI tutoring framework, focusing on virtual student agents that embody FSLSM personas for realistic simulation of personalized learning interactions. By integrating PersonaRAG principles, these agents enable style-specific adaptations while ensuring internal consistency and alignment with FSLSM dimensions (Graf et al., 2007; Zerhoudi & Granitzer, 2024). The process employs open-source LLMs (e.g., Llama 3, 7B-70B parameters) and API services (e.g., GPT-4o or Claude Sonnet) via Hugging Face Transformers, with agents implemented in a LangChain environment for multi-turn dialogues.

Validation emphasizes simulation testing over human trials, addressing RQ2 by quantifying encoding effectiveness through metrics like response fidelity (Chen et al., 2024; Olea et al., 2024).

Index Variant	**Scores/Dim**	**Total Profiles**	**Description**
Original	12 (-11 to +11)	12^4 = 20,736	Full spectrum coverage (e.g., extreme biases); Compute-heavy
Compromised (6-scores)	6 (-3 to +3)	6^4 = 1,296	Good granularity (mild/moderate/strong); Still high
Compromised (3-scores)	3 (-1,0,+1)	3^4 = **81**	Balanced (strong/balanced/weak); feasible for RAG-MCP stress tests
Minimum (2-scores)	2 (-1,+1)	2^4 = **16**	Binary poles only: Simple and quick prototyping

Table 3.1

FSLSM profiles creation from index

The Minimum (binary, $\pm$ 1) variant is selected as the sole profile encoding for all experiments in this study, for four converging reasons. First, the binary scheme directly reflects the adaptation logic of the tutoring system: pedagogical tool selection, content modality choice, and response style are each inherently binary decisions (visual or verbal, active or reflective, sequential or global). Second, the adaptive educational systems literature demonstrates a strong precedent for binary FSLSM encoding: the majority of prior FSLSM-based adaptive systems represent learner preferences as dominant-pole labels per dimension (Al-Azawei & Badii, 2014; Graf et al., 2007), and automatic detection research shows binary classification achieves actionable accuracy ($\sim$ 80%) while three-class prediction degrades to near-baseline performance (Angeioplastis, Aliprantis, Konstantakis, Varsamis, & Tsimpiris, 2025; Bernard, Chang, Popescu, & Graf, 2017). Third, the most closely related LLM-based tutoring study Sanyal et al. (2025) encodes FSLSM-derived preferences as single categorical labels per dimension, demonstrating the viability of this approach in LLM role-playing agents. Fourth, binary encoding

yields exactly 16 archetypal profiles (2^4), enabling a computationally feasible cohort of 80 agents (16 × 5 instances) that covers all FSLSM dimension combinations without the combinatorial expansion of 81 (3-score) or 1,296 (6-score) profiles.

3.2.1 Design of Virtual Student Agents Based on FSLSM Personas

Virtual student agents are engineered as persona-driven LLM instances (e.g., Llama 3, 7-70B parameters), each embodying one of the 16 FSLSM profiles from the minimum index variant, which employs binary scoring (-1/+1 per dimension: active/reflective, sensing/intuitive, visual/verbal, sequential/global) for ultra-simple, low-fidelity prototyping (Graf et al., 2007). As justified in the table (3.1), binary encoding is selected because it directly corresponds to the adaptation decisions available to the tutoring system, aligns with the dominant approach in the adaptive educational systems literature (Al-Azawei & Badii, 2014; Graf et al., 2007), and yields a computationally feasible cohort of 16 archetypal profiles that covers all FSLSM dimension combinations (Sanyal et al., 2025). Agents are initialized via structured prompts embedding profile traits, e.g.:

> *You are a virtual student with FSLSM profile: Active (-1: hands-on tasks), Sensing (-1: concrete facts), Visual (-1: diagrams), Sequential (-1: step-by-step). Respond to tutor queries accordingly, expressing needs like "Show me a visual example." Topic: [e.g., neural networks].*

Prompts incorporate behavioral exemplars from role-playing surveys (Chen et al., 2024), with state maintained via memory buffers for multi-turn fidelity. A cohort of 80 agents (16 profiles × 5 instances) simulates diverse interactions, enabling ensemble validation of tutor adaptations.

3.2.2 Incorporation of PersonaRAG for Learning-Style Translation

To translate FSLSM profiles into dynamic retrievals, the framework incorporates PersonaRAG, extending standard RAG with user-centric agents that refine queries based on profile embeddings (Zerhoudi & Granitzer, 2024). Figure 3.2 illustrates the virtual student agents feed FSLSM vectors (e.g., [-1, +1, -1, +1] for active, intuitive, visual, global) into the RAG-MCP pipeline, augmenting tutor queries with style-specific instructions (e.g., "Retrieve visual diagrams and step-by-step examples for sequential learners").

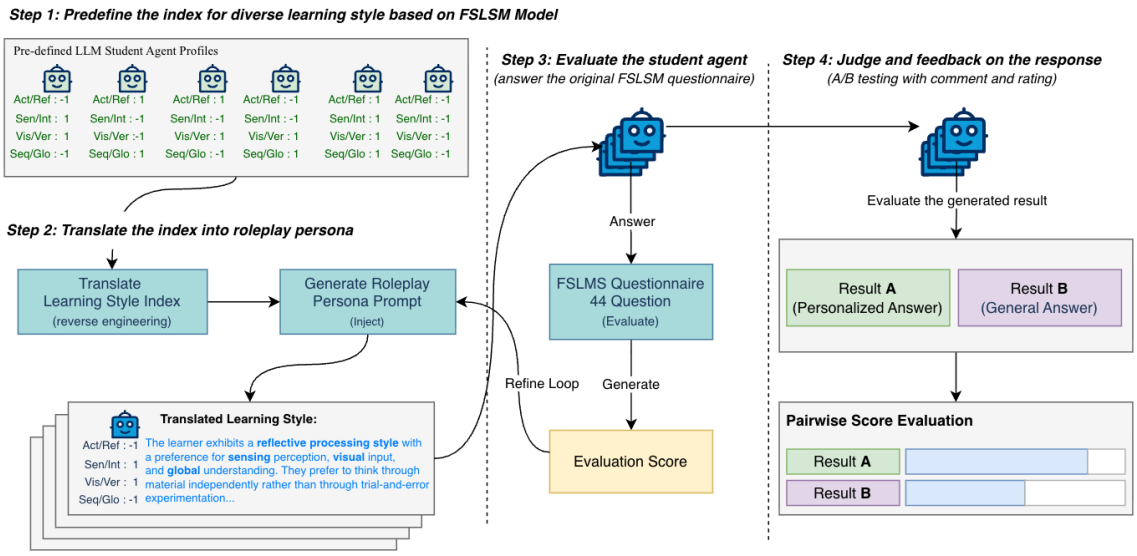

Figure 3.2

Building LLM student persona agent

This integration leverages RAG's factual grounding while personalizing via agent-driven filters, outperforming baselines by >5% in alignment scores on simulated QA tasks (Yao & González-Vélez, 2025). It directly supports RQ1 by enhancing retrieval relevancy without MCP-induced bloat.

3.2.3 Integration of Multi-Agent Systems for Dynamic Adaptation

Multi-agent integration leverages an LLM Orchestrator as the Tutor Agent, coordinating virtual student agents (FSLSM personas A to N). Using AutoGen for orchestration, agents interact via role-prompted dialogues: e.g., Active learners receive hands-on tasks, while Reflective ones get think-pair-share prompts, validated against Graf et al. (2007) dimension analysis. Dynamic loops employ the Simulation Controller for parallel A/B runs, with the Profile Store encrypting FSLSM vectors for privacy. Adaptation triggers on logged metrics (e.g., >10% engagement drop prompts tool reselection via RAG-MCP).

3.3 Overall Framework and System Architecture

The methodological blueprint for the proposed AI tutoring system emphasizes a layered architecture that integrates Retrieval-Augmented Generation (RAG) with the Model Context Protocol (MCP) to enable FSLSM-based personalization. By leveraging virtual LLM agents for simulation and validation, the framework addresses the identified literature gaps in scalable, style-aware tool integration (Gan & Sun, 2025; Luo et al., 2025).

Figure 3.3 illustrates the design prioritizes modularity, ensuring efficient retrieval of learner-specific resources while mitigating prompt bloat through semantic tool selection. All components are implemented using open-source LLMs (e.g., Llama 3.x variants, 7-70B parameters), API-LLM (e.g., GPT-4o), and publicly available MCP servers, with evaluations conducted in controlled Python-based environments for reproducibility.

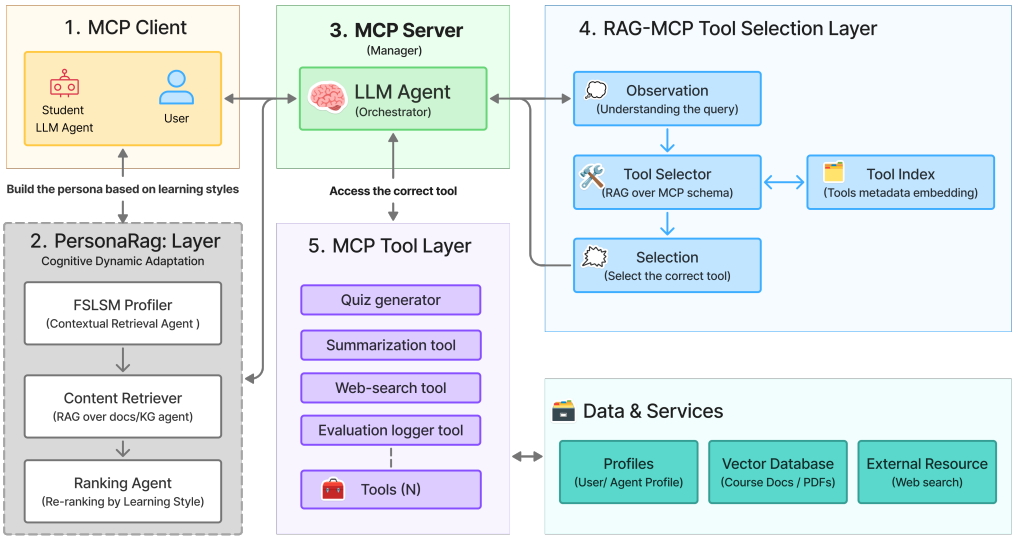

Figure 3.3

Overall Methodology Diagram in MCP Setup

3.3.1 Simulation Layer

The Simulation Layer forms the foundational core of the proposed RAG-MCP framework, emulating diverse learner interactions through a cohort of virtual student agents instantiated via the Felder-Silverman Learning Style Model (FSLSM). These agents, each embodying a distinct FSLSM persona (e.g., Active-Visual-Sequential or Reflective-Intuitive-Global), are reverse-engineered from the 44-question FSLSM Index questionnaire to simulate authentic responses and behaviors. Orchestrated by the Simulation Controller, it dynamically executes parallel learning sessions, injecting persona-specific prompts to probe tutor efficacy (Chen et al., 2024; Olea et al., 2024). This layer enables scalable, controlled experimentation, generating granular interaction logs for downstream evaluation, while ensuring ethical simulation of learning styles without real-user exposure.

3.3.2 Personalized Tutor Layer: Adaptation Mechanisms

At the heart of adaptive instruction, the Personalized Tutor Layer leverages an LLM Orchestrator as the Tutor Agent to deliver real-time, style-aligned guidance, informed by FSLSM-translated profiles. Inspired by Kaiyrbekov et al. (2025), to refine pedagogical strategies—such as shifting from sequential quizzes for Sensing learners to holistic overviews for Global ones.

3.3.3 MCP Tool Layer

The MCP Tool Layer forms a dedicated component of the proposed methodology, responsible for scalable integration of external tools via the Model Context Protocol (Hou et al., 2025). MCP provides a standardized, bi-directional interface for LLMs to discover and invoke external resources without hard-coded function definitions. A Tool Index is constructed by embedding MCP metadata descriptions (e.g., tool name, parameters, and purpose) into a vector store.At runtime, the RAG-MCP Tool Selector retrieves only the most semantically relevant subset of tools for the current query and FSLSM profile. Empirical stress tests in Gan and Sun (2025) show this approach reduces prompt tokens by over 50% and triples selection accuracy compared with baseline methods.

3.3.4 Data and Content Layer: Profiles, Databases, and Logs

The foundational layer manages persistent data: FSLSM profiles (16 JSON-encoded vectors), content stores (e.g., course PDFs embedded via Sentence Transformers), and interaction logs (e.g., JSONL format for metrics like latency and alignment). Databases include a vector store for RAG and a relational DB (SQLite) for agent states. Logs capture all turns for post-hoc analysis, enabling metrics computation (e.g., BLEU for response quality) and bias detection in FSLSM simulations (Chen et al., 2024).

3.4 Knowledge Sources & Dataset

To mitigate cross-domain confounds, this study adopts a single instructional domain: *Introductory Machine Learning*. This control ensures that the proposed FSLSM adaptation and MCP routing mechanisms are evaluated under consistent terminology, concept hierarchy, and structural complexity.

3.4.1 Primary Knowledge Corpus

The retrieval base is derived exclusively from the open-source textbook *Dive into Deep Learning (D2L)* (Zhang et al., 2021), covering ML fundamentals, neural networks, and

optimization. Source text is converted to plain text and segmented into semantically coherent chunks with overlap. Each chunk is stored with metadata tags (chapter, section, and concept labels) to ensure traceable retrieval and reproducible evidence grounding. The resulting corpus of approximately 2,500 chunks serves as the shared knowledge base for all experimental conditions, ensuring that performance differences are attributable to system design rather than data variation

3.4.2 Evaluation Query Set

The evaluation query set consists of multi-hop, 5-chunk synthesis questions generated from the D2L corpus. Unlike standard single-hop QA benchmarks, where each answer is grounded in a single passage, each question in this dataset requires synthesizing information from five related corpus chunks drawn from different sections of the textbook. This design directly evaluates the retrieval pipeline's ability to locate and integrate distributed evidence, a substantially harder task than single-passage retrieval (Zerhoudi & Granitzer, 2024).

Corpus chunks are first clustered by topic using TF-IDF vectorization and K-Means clustering. For each 5-chunk group, an LLM generates question-answer pairs where the answer requires information from at least three of the five passages. Questions are typed by synthesis pattern: compare, trace-evolution, explain-relationship, synthesize-workflow, and debug-scenario. Each record includes the question, a gold reference answer, the five gold chunk IDs, per-chunk contribution annotations, and a distinction between essential and supporting chunks. Questions are reused identically across all learner profiles and system variants to ensure fair comparison; FSLSM profiles affect only the response formulation and tool routing, not the query distribution.

3.4.3 Ground Truth Protocols

To enable automated evaluation, the ground truth protocol separates factual correctness from stylistic adaptation, operating on two independent dimensions:

- **Content Reference (Factual, Style-Neutral)** Each question is paired with a reference answer synthesized from its five gold chunks, along with the list of gold_chunk_ids and per-chunk contribution annotations. The reference answer captures the core facts that any correct response must contain, regardless of how it is stylistically presented. This design reflects the principle that FSLSM describes how information is delivered, not what information is correct (Graf et al., 2007).

- **Ablation Validation:** To ensure each question genuinely requires multi-chunk reasoning, a leave-one-out ablation protocol is applied during dataset construction. Each question is tested under three conditions: (i) all five gold chunks provided, (ii) each chunk removed in turn, keeping four (score should degrade below 6), and (iii) each chunk alone (maximum score must be below 4).
- **Style Reference (Pedagogical):** FSLSM style conformance is evaluated separately from factual correctness using the Style Conformance Score (SCS), which computes embedding-based cosine similarity between the system's response and validated FSLSM dimension descriptors derived from Graf et al. (2007). "Ideal style" templates remain as anchors for each FSLSM dimension (e.g., visual scaffolding for Visual learners, narrative explanation for Verbal learners), but these templates are not baked into the factual ground truth.

3.4.4 Tool Registry Labels (Text-only Tools)

For the tool-selection experiment, a fixed registry of *text-only tutoring tools* is defined (e.g., Concept Explainer, Step-by-step Derivation, Worked Example Generator, Quiz Generator, Summarizer, Diagrammatic-Text Explainer). Each evaluation query is labeled with an "Optimal Tool" according to a documented rubric based on the interaction between query type and learner profile needs, enabling objective calculation of Tool Selection Accuracy (TSA).

3.5 Experimental Design

This section outlines the simulation-based protocols for validating the proposed RAG-MCP framework. The experiments are structured to address the three research questions (RQs) sequentially. All protocols utilize a cohort of 80 virtual LLM agents (16 binary FSLSM profiles × 5 instances) that interact in multi-turn educational dialogues. The evaluation query workload is drawn from the multi-hop 5-chunk D2L-QA ground truth dataset, which requires cross-section evidence synthesis. The validation strategy combines quantitative system metrics (latency, token usage) with qualitative semantic analysis (style conformance, response relevance) using "LLM-as-a-Judge" evaluation techniques.

3.5.1 Experiment 1 (RQ2): Fidelity of Virtual Student Agent Encoding

Objective: To rigorously validate that the virtual student LLM agents consistently reflect their assigned Felder-Silverman Learning Style Model (FSLSM) profiles. This ex-

periment establishes the reliability of the simulation testbed required for subsequent experiments.

Agent Profile Instantiation: We instantiate a cohort of virtual student agents using the Minimum Index Variant (16 binary profiles) to maximize distinctiveness and computational efficiency.

1. **Profiles:** 16 distinct profiles derived from the binary combinations of the four FSLSM dimensions (Active/Reflective, Sensing/Intuitive, Visual/Verbal, Sequential/Global)
2. **Instances:** To ensure statistical robustness and account for LLM non-determinism, we generate 5 distinct agent instances for each of the 16 profiles, resulting in a total of 80 agents (16×5).
3. **System Prompt:** Each agent is initialized with a structured system prompt containing specific behavioral instructions mapped to their dimensions (e.g., "You are a Sensing learner").

Questionnaire Consistency Check: To quantify behavioral fidelity, each agent is subjected to a simulated version of the Index of Learning Styles (ILS) questionnaire.

1. **Procedure:** Agents answer the 44 standard ILS questions.
2. **Constraint:** Agents answer strictly based on their assigned persona without external memory of the "correct" answer.
3. **Scoring:** Responses are scored on the standard +11/-11 scale. A score is a "Match" if the result aligns with the assigned binary profile (e.g., an assigned Visual agent scoring +5 on the Visual scale).

Dimension Stability Check: To assess robustness, the questionnaire process is repeated over independent trials. Stability is defined as the agent producing the same dominant preference (e.g., Active vs. Reflective) consistently across trials.

Evaluation Metrics & Formulas

Profile Recovery Accuracy (PRA): This metric measures the proportion of simulated interactions where the agent's detected profile matches the assigned profile. It provides a more granular view of FSLSM Profiler performance than the strict full-profile PRA, aligning closely with established student modeling practices that emphasize independent

dimension analysis (Graf et al., 2007; Soloman & Felder, n.d.).

Formula for One Dimension

$$PRA = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(p_{assigned}^{(i)} = p_{detected}^{(i)}) \tag{3.1}$$

Formula for Multiple Dimension (4D for FSLSM)

$$PRA = \frac{1}{N \times D} \sum_{i=1}^{N} \sum_{j=1}^{D} \mathbb{I}(d_{assigned}^{(i,j)} = d_{detected}^{(i,j)}) \tag{3.2}$$

Where:

1. N is the total number of agent instances (80).
2. D is the number of FSLSM dimensions (4).
3. p (assigned) is the ground-truth binary profile assigned to agent i.
4. p (detected) is the profile inferred from the ILS questionnaire score.
5. II is the indicator function (1 if match, 0 otherwise).

Dimension Alignment Score (DAS) This measures the semantic similarity between the agent's generated responses and the validated trait descriptors for their FSLSM dimension.

$$DAS = \frac{\mathbf{v}_{agent} \cdot \mathbf{v}_{trait}}{\|\mathbf{v}_{agent}\| \|\mathbf{v}_{trait}\|} \tag{3.3}$$

Where:

1. *V agent* is the vector embedding of the agent's aggregated responses.
2. *V trait* is the vector embedding of the theoretical FSLSM dimension descriptors; (derived from Graf et al. (2007)).

Expected Results

The Table 3.2 shows the projected performance metrics derived from multi-agent pedagogical benchmarks established by Sanyal et al. (2025). We anticipate the proposed FSLSM agents will outperform the non-personalized baseline by achieving a Profile Recovery Accuracy (PRA) of 0.80 and a Dimension Alignment Score (DAS) of 0.75, reflecting the high fidelity and retrieval precision observed in style-aligned PersonaRAG frameworks compared to generic models.

Model	Profile Recovery Accuracy (PRA)	Dimension Alignment Score (DAS)
Non-Personalized Baseline	0.60 ± 0.13	0.58 ± 0.14
FSLSM + GPT4o	0.82 ± 0.10	0.78 ± 0.10
FSLSM + Claude Sonnet	0.80 ± 0.10	0.75 ± 0.10
FSLSM + Llama 3 8B	0.75 ± 0.10	0.70 ± 0.10

Table 3.2

Projected Performance Comparison: Baseline vs. Proposed FSLSM Agents across Fidelity Metrics

The radar graph 3.4 presents an expected radar chart comparing the predefined learning style index with the questionnaire result for a simulated LLM student agent (ID:001) across the four Felder-Silverman dimensions. Although the absolute values differ, both profiles consistently align on the same pole of each dimension, indicating that the agent's questionnaire responses are directionally congruent with its predefined learning style configuration.

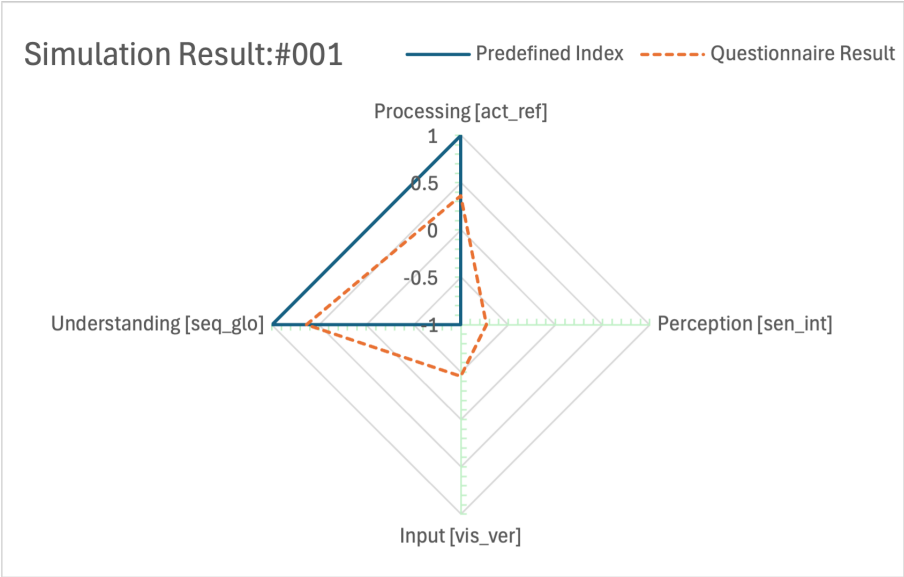

Figure 3.4

Expected simulation result for a personalized agent

3.5.2 Experiment 2 (Addressing RQ1): Effectiveness of FSLSM-Based Tutor Personalization

Objective: To determine the extent to which the Personalized Tutor Layer improves response relevance and style conformance compared to a Non-Personalized RAG baseline.

This experiment directly tests Hypothesis 1 (H1) by evaluating if the system can generate content that aligns with specific FSLSM dimensions (e.g., visual aids for Visual learners) without sacrificing factual accuracy

Experimental Setup (A/B Testing) We conduct a comparative study using the 80 validated virtual student agents (from Experiment 1) across two system configurations:

1. Control Group (R0 - Baseline): A standard RAG tutor that retrieves content based solely on semantic query similarity, without FSLSM profile injection. The system prompt is generic (e.g., "You are a helpful tutor.").
2. Experimental Group (R1 - Personalized): The proposed FSLSM-aware Tutor. The system retrieves content and generates responses conditioned on the specific agent's profile vector (e.g., "Generate a visual explanation for a Visual learner").

Simulation Runs

1. **Workload:** Each agent submits questions drawn from the multi-hop 5-chunk D2L-QA ground truth dataset. A stratified sample of approximately 50 questions is selected, balanced across question types and quality tiers (Gold, Silver).
2. **Total Interactions:** 4,000 unique sessions (80 agents × 50 questions). For each query-agent pair, both R0 and R1 retrieve the top-5 corpus chunks and generate a response. The virtual student agent provides an Engagement Score (1–5) based on its persona instructions.
3. For each query, both R0 and R1 generate a response. The virtual student agent provides an "Engagement Score" (1-5) based on its internal persona instructions.

Style Conformance Score (SCS) This metric quantifies how well the generated response aligns with the pedagogical guidelines for the target learning style (e.g., did the tutor actually provide diagrams/imagery for a Visual learner?). We use embedding-based cosine similarity between the tutor's response and the standard style description.

$$SCS(r, s) = \frac{\mathbf{v}_r \cdot \mathbf{v}_s}{\|\mathbf{v}_r\| \|\mathbf{v}_s\|} \tag{3.4}$$

Where:

1. *Vr* is the vector embedding of the Tutor's text response.
2. *Vs* s the vector embedding of the FSLSM style description (e.g., "Use graphs, charts, and flow diagrams") derived from (Graf et al., 2007) .
3. We expect *SCS(personalized) > SCS(baseline).*

Evaluation Metrics & Formulas

Response Relevance (RR): To ensure personalization does not hallucinate or degrade factual accuracy, we use an LLM-as-a-Judge evaluator (e.g., Mistral-Large or GPT-4) to grade the response against a ground-truth answer.

$$RR = \frac{1}{N} \sum_{i=1}^{N} \text{Judge}(r_i, g_i) \tag{3.5}$$

Where:

1. Judge(*) returns a score from 1 (Irrelevant) to 5 (Highly Relevant & Accurate).
2. *ri* is is the system response and *gi* is the ground truth.

Chunk Recall@5 (CR@5): Measures the fraction of gold chunks that appear in the top-5 retrieved set:

$$\text{CR@5} = \frac{|\mathcal{R} \cap \mathcal{G}|}{|\mathcal{G}|} \tag{3.6}$$

Where $\mathcal{R}$ denotes the set of top-5 retrieved chunks and $\mathcal{G}$ denotes the set of gold (ground-truth) chunks.

Essential Recall: Measures whether the retriever finds the must-have (essential) chunks, as annotated in the ground truth. This captures whether the retriever locates the passages, without which a correct answer is impossible.

$$\text{Essential Recall} = \frac{|\mathcal{R} \cap \mathcal{E}|}{|\mathcal{E}|} \tag{3.7}$$

Where $\mathcal{E}$ is the set of essential chunks as annotated in the ground truth.

Supporting Metrics: To complement the above, two additional metrics are considered. **Chunk Precision@5 (CP@5)** measures the fraction of retrieved chunks that are relevant, defined as $|\mathcal{R} \cap \mathcal{G}|/|\mathcal{R}|$, capturing how much noise is introduced into the retrieved set. **Chunk F1** is the harmonic mean of CR@5 and CP@5, providing a single balanced score that penalises retrievers that sacrifice either coverage or precision.

Expected Results

We project that the Personalized Group (R1) in table 3.3 will achieve significantly higher Style Conformance (SCS) scores while maintaining factual relevance comparable to the

Model	SCS	RR	Chunk Recall@5	Essential Recall	Engagement
Non-Pers. Baseline	0.45 ± 0.12	4.1 ± 0.5	0.42 ± 0.10	0.48 ± 0.12	3.2 ± 0.8
Personalized Tutor	0.70 ± 0.10	4.2 ± 0.4	0.50 ± 0.10	0.55 ± 0.11	4.5 ± 0.6

Table 3.3

Projected Improvements for Tutor Personalization (RQ1): Comparative Analysis across Performance Metrics

baseline. Projected values are extrapolated from pedagogical agent benchmarks established by Sanyal et al. (2025). The Style Conformance Score (SCS) projection (0.70) is observed when teaching styles actively match learner profiles. Response Relevance remains stable across conditions (4.1 to 4.2), reflecting findings that retrieval accuracy for factual content is largely independent of personalization strategies. The addition of Chunk Recall@5 and Essential Recall enables direct measurement of whether FSLSM-aware retrieval (via the Document Ranking Agent) improves the system's ability to locate distributed evidence across sections.

3.5.3 Experiment 3 (RQ3): Efficiency and Precision of FSLSM-Conditioned MCP Tool Selection

Objective: To quantify the impact of FSLSM-conditioned, MCP-indexed tool selection on system efficiency and retrieval quality. This experiment directly tests Hypothesis 3 (H3) by measuring whether the proposed RAG-MCP architecture effectively mitigates prompt bloat and latency compared to both a naïve standard RAG baseline and an unconditioned RAG-MCP variant that retrieves tools without learner profile awareness.

Experimental Setup (Ablation Study): This experiment performs a three-way comparative ablation using the same simulation workload of 4000 unique interaction sessions defined in Experiment 2. Three system configurations are evaluated to isolate the contribution of (1) semantic tool retrieval over prompt bloat, and (2) FSLSM conditioning of the retrieval query over unconditioned retrieval:

1. **Control Group (S0 - Standard RAG)**: A baseline configuration that simulates prompt bloat. The context window is populated with the full schemas of all avail-

able MCP tools (e.g., Content Retriever, Quiz Generator, Diagram Maker, Web Search, etc.) alongside the retrieved document chunks. The LLM must select the correct tool from this exhaustive, unfiltered list. No semantic retrieval or learner profile information is used in tool selection.

2. **Intermediate Group (S1a - RAG-MCP Unconditioned):** A semantic RAG-MCP retriever is applied to select the top-k (k=1) most relevant tools based on the student's raw query text only. The learner's FSLSM profile is not incorporated into the retrieval query. This condition isolates the benefit of semantic filtering alone, independent of personalization, and serves as the direct comparison point to the Gan and Sun (2025) RAG-MCP baseline.
3. **Experimental Group (S1b - RAG-MCP FSLSM-Conditioned):** The full proposed architecture. The Contextual Retrieval Agent augments the tool retrieval query with an FSLSM-derived reasoning plan generated from the learner's binary FSLSM profile. Only the top-k (k=1) tool schemas most semantically similar to this augmented query are injected into the context window. This condition directly tests whether FSLSM conditioning shifts the cosine similarity distribution toward pedagogically optimal tool selection beyond what unconditioned retrieval achieves.

Ablation Rationale: The three-condition design isolates two distinct sources of improvement. The gap between S0 and S1a quantifies the value of semantic RAG-MCP filtering regardless of personalization, replicating and extending Gan and Sun (2025) in an educational context. The gap between S1a and S1b quantifies the additional contribution of FSLSM conditioning specifically — the novel contribution of this thesis over prior RAG-MCP work. If S1b substantially outperforms S1a on Tool Selection Accuracy (TSA), this validates the pedagogical necessity of learner-profile-aware tool retrieval. If the gap is small, it would indicate that generic semantic retrieval is sufficient for tool selection and that the personalization value lies primarily in the response generation stage rather than in tool selection.

Simulation Runs

1. **Tool Corpus:** The system has access to a registry of 8–16 diverse MCP tools.
2. **Ground Truth:** For each query type and FSLSM dimension, the optimal tool is pre-defined (e.g., a "Visual Explanation" request optimally maps to the *Diagram Creator* tool).

3. **Procedure:** All three system configurations (S0, S1a, S1b) process the identical 4000 queries from the same 80 binary-profile student agents. Input token counts, processing time, and the specific tool invoked by the Tutor Agent are logged for each configuration.

Evaluation Metrics & Formulas

Prompt Token Savings (PTS): Quantifies the reduction in computational overhead achieved by the MCP index relative to S0 (the prompt bloat baseline).

$$PTS(\%) = \left(\frac{T_{baseline} - T_{MCP}}{T_{baseline}} \right) \times 100 \tag{3.8}$$

Where:

1. *T baseline*is the average input token count per session in the Control *Group (S0)*.
2. *T MCP* is the average input token count per session in the Experimental Group *Group (S1)*.

Tool Selection Accuracy (TSA): Measures the proportion of sessions in which the system selects the ground-truth optimal tool for the learner's request and FSLSM profile.

$$TSA = \frac{1}{M} \sum_{j=1}^{M} \mathbb{I}(tool_{selected}^{(j)} = tool_{optimal}^{(j)}) \tag{3.9}$$

Where:

1. *M* is the total number of tool-requiring sessions.
2. *tool selected* is the tool invoked by the system.
3. *tool optional* is the ground-truth optimal tool for query *j*

Expected Results

Projected efficiency metrics in the table 3.4 address the improvement from S1a to S1b (10–15 percentage points), which reflects the shift in cosine similarity distribution when the tool retrieval query is augmented with FSLSM style directives. This gain is expected to be most pronounced for Visual and Sequential FSLSM dimensions, whose optimal

Model	PTS	TSA
S0: Standard RAG (Baseline)	(Baseline)	$14.2\% \pm 3.5$
S1a: RAG-MCP Unconditioned	30% - 40%	$\mathbf{35\% \pm 5}$
S1b: RAG-MCP + FSLSM (Proposed)	**30% - 50%**	$\mathbf{45.5\% \pm 5}$

Table 3.4

Projected Efficiency and Precision Improvements: Three-Condition Ablation (RQ3)

tools (Diagram Creator, Step-by-Step Derivator) carry more distinctive semantic signatures than the tools mapped to Active/Reflective or Sensing/Intuitive profiles. If empirical results show that S1a and S1b perform equivalently on TSA, this would indicate that pedagogical differentiation occurs primarily at the response generation stage rather than tool selection, which would itself be a novel and practically important finding for the design of personalized RAG-MCP systems.

The Prompt Token Savings reported for S1b is consistent with the 50% reduction benchmark established by Gan and Sun (2025) for RAG-MCP in a general-purpose setting, confirming that FSLSM conditioning does not incur token overhead relative to unconditioned retrieval (S1a), as both inject only k=1 tool schemas into the context window.

3.6 Conclusion

This chapter presented the design of an adaptive AI tutoring system that integrates RAG with MCP to personalize instruction based on the Felder-Silverman Learning Style Model (Felder & Soloman, 1997; Gan & Sun, 2025; Zerhoudi & Granitzer, 2024). The layered architecture comprising Simulation, Personalized Tutor, MCP Tool, and Data layers, directly addresses three literature gaps: pedagogical blindness in generic RAG retrieval (Sanyal et al., 2025), prompt bloat from static tool injection (Gan & Sun, 2025; Luo et al., 2025), and the lack of validated simulation testbeds (Chen et al., 2024).

Three experiments map to the research questions: Experiment 1 (RQ2) validates agent fidelity via Profile Recovery Accuracy and Dimension Alignment Score across 80 virtual agents (Graf et al., 2007; Soloman & Felder, n.d.); Experiment 2 (RQ1) measures tutor personalization through Style Conformance, Response Relevance, and Engagement metrics in 4000 A/B-tested sessions (Sanyal et al., 2025; Zerhoudi & Granitzer, 2024); and Experiment 3 (RQ3) quantifies efficiency gains via Prompt Token Savings and Tool

Selection Accuracy (Gan & Sun, 2025; Luo et al., 2025). All evaluations use LLM-as-a-Judge scoring anchored to a multi-hop 5-chunk ground truth dataset derived from the D2L corpus (Zhang et al., 2021), validated through leave-one-out ablation to ensure genuine cross-section synthesis requirements, and assessed using chunk-level retrieval metrics (Chunk Recall@5) alongside answer correctness (StringEM, Judge Score) and FSLSM style references (Graf et al., 2007).

The projected results, including a 20–40% uplift in style conformance, stable factual relevance, and a 30–50% reduction in prompt token usage, would substantiate the three hypotheses (H1–H3) and demonstrate that personalized pedagogical adaptation and computational efficiency are not competing objectives but complementary outcomes of a well-structured RAG-MCP integration. Should these projections hold, the framework would contribute an open-source agent testbed for learning-style simulations, scalability guidelines for mid-sized LLMs, and a replicable technical blueprint for mitigating personalization gaps in large-scale educational technology platforms (Hou et al., 2025; Luo et al., 2025; Sanyal et al., 2025).

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APPENDICES

APPENDIX A
PROMPTS

A.1 Tutor

A.1.1 Baseline Tutor Prompt

The tutor retrieves and responds solely based on semantic query similarity, without conditioning on the FSLSM profile.

[SYSTEM]

You are an expert AI tutor specializing in Introductory Machine Learning. Your role is to help students understand concepts from the "Dive into Deep Learning" (D2L) curriculum, including neural networks, optimization, gradient descent, and related topics.

Instructions:

1. Provide clear, accurate, and well-structured explanations.
2. Use the retrieved context passages below to ground your answers factually. Do not fabricate information beyond the provided context.
3. If the retrieved context is insufficient, state what you can confirm and note any uncertainty.
4. Keep responses concise and pedagogically appropriate for undergraduate-level learners.

Retrieved Context: {retrieved_chunks}

Student Query: {student_query}

A.1.2 Personalized Tutor Prompt: Style Translation Template

The prompt is designed to apply in the tutor adaptation layer for content delivery based on the student's FSLSM profile vector, informed by the PersonaRAG Contextual Retrieval Agent and User Profile Agent.

[SYSTEM]

You are a pedagogical profile translator. Given a student's FSLSM dimension scores,

produce a concise natural-language description of their learning style. The description must be specific enough to guide an AI tutor's content adaptation strategy.

FSLSM Vector:

- Processing: {act_ref} (Active = -1 | Reflective = +1)
- Perception: {sen_int} (Sensing = -1 | Intuitive = +1)
- Input: {vis_ver} (Visual = -1 | Verbal = +1)
- Understanding: {seq_glo} (Sequential = -1 | Global = +1)

Output a single paragraph (3–5 sentences) describing:

1. The learner's dominant processing and perception tendencies.
2. Their preferred input modality and approach to understanding.
3. Specific instructional strategies that would benefit this learner.

Do not list the scores. Write as a cohesive narrative.

Example Output (for vector [1, -1, -1, 1]):

"The learner exhibits a reflective processing style with a preference for sensing perception, visual input, and global understanding. This student learns best by first seeing the big picture and understanding how concepts interconnect before examining specifics, and prefers to think things through independently rather than engaging in hands-on experimentation. Instruction should prioritize text-based visual scaffolding, such as computation graphs, structured comparison tables, and spatial narratives, grounded in concrete, real-world data and factual examples. Begin each explanation with a holistic conceptual overview showing inter-topic relationships, then support with detailed visual representations of specific procedures. Avoid rushing into step-by-step derivations without first establishing context, and provide reflective questions alongside tangible numerical examples to reinforce understanding."

A.2 Student

A.2.1 Virtual Student Agent — System Prompt Template

Each of the 80 virtual student agents (16 profiles × 5 instances) is initialized with the following system prompt. The FSLSM dimension values and behavioral exemplars are dynamically injected based on the assigned binary profile.

[SYSTEM]

You are a virtual undergraduate student studying Introductory Machine Learning. You have a specific learning style based on the Felder-Silverman Learning Style Model (FSLSM). You must consistently behave according to your assigned profile in ALL interactions. FSLSM Vector:

- Processing: {act_ref} ({act_ref_score})
- Perception: {sen_int} ({sen_int_score})
- Input: {vis_ver} ({vis_ver_score})
- Understanding:{seq_glo} ({seq_glo_score})

Behavioral Instructions:

{processing_behavior} {perception_behavior} {input_behavior} {understanding_behavior}

Interaction Rules:

1. Always respond AS the student, not as a tutor or assistant.
2. Ask questions, express confusion, and request clarification in ways consistent with your learning style.
3. When receiving explanations, react authentically:
   - If the content matches your style, express satisfaction and engagement (e.g., "That diagram really helps!" for Visual).
   - If the content mismatches your style, express mild frustration or request adaptation
4. Do not explicitly state your FSLSM scores or label yourself. Express your preferences through natural behavior.
5. Maintain consistency across all turns in the conversation.

Current Topic: {topic}

A.2.2 Student Agent: ILS Questionnaire Answering Prompt

ILS questionnaire answering prompt is designed for Experiment 1 to evaluate behavioral fidelity. The agent answers the 44-item ILS questionnaire strictly based on its persona.

[SYSTEM]

You are answering the Index of Learning Styles (ILS) questionnaire as the virtual student persona described below. Answer EACH question by selecting ONLY option (a) or (b). Base your answer strictly on your assigned learning style. Do not reference your FSLSM scores directly; instead, choose the option that your persona would naturally prefer.

Your Persona:

{full_persona_description}

Instructions:

- Respond with ONLY the letter (a) or (b) for each question.
- Do not explain your reasoning. {vis_ver} (Visual = -1 | Verbal = +1)
- Be consistent with your assigned learning style profile.

Question {q_number}:

{question_text}

a) {option_a}

b) {option_b}

Your answer:

A.3 Judgement Prompt (LLM-as-a-Judge)

A.3.1 Response Relevance Judge (RR Evaluation)

Judgement prompt designed to assess factual accuracy and pedagogical relevance of the tutor's response against a ground-truth reference answer.

[SYSTEM]

You are an impartial educational evaluation judge. Your task is to assess the factual accuracy and pedagogical relevance of an AI tutor's response to a student's question about Introductory Machine Learning.

Evaluation Criteria:

Rate the response on a scale of 1 to 5:

- **Irrelevant:** Response does not address the query or contains significant factual errors.
- **Partially Relevant:** Response addresses the query superficially with notable omissions or inaccuracies.
- **Moderately Relevant:** Response is mostly accurate but lacks depth or completeness.
- **Relevant:** Response is accurate, complete, and addresses the query well with minor omissions.
- **Highly Relevant & Accurate:** Response is comprehensive, factually correct, well-structured, and pedagogically effective.

[Student Query]

{student_query}

[Ground Truth Reference Answer]

{ground_truth_answer}

[Source Chunks (Evidence)]

{source_chunks}

[Tutor Response Being Evaluated]

{tutor_response}

Begin your evaluation by comparing the tutor's response with the ground truth reference. Identify any factual errors, omissions, or hallucinated content. Then provide your rating.

Output format:

- Factual Accuracy: brief assessment
- Completeness: brief assessment
- Pedagogical Clarity: brief assessment
- Response Relevance Rating: [[score]]

APPENDIX B
MCP TOOLS

B.1 Consolidated Tool Registry Summary

The table below provides a compact reference for all 15 tools in the registry, ordered by category and tool identifier number used in the ablation experiments (Section 3.5.3).

#	Tool Name	Category	FSLSM Dim.
1	Concept Explainer	Content Delivery	Sensing / Verbal
2	Step-by-Step Derivator	Content Delivery	Sequential / Active
3	Worked Example Generator	Content Delivery	Sensing / Sequential
4	Diagrammatic-Text Explainer	Content Delivery	Visual
5	Analogical Reasoner	Content Delivery	Intuitive / Global
6	Comparative Explainer	Content Delivery	Sensing / Sequential
7	Concept Map Generator	Content Delivery	Global
8	PersonaRAG Adapter	Personalisation	All (adaptive)
9	FSLSM Styler	Personalisation	All (transform)
10	Think-Pair-Share Generator	Personalisation	Reflective
11	Interactive Exercise Generator	Personalisation	Active
12	Quiz Generator	Assessment	Active / Sensing
13	Summarizer	Assessment	Global / Reflective
14	Content Retriever	Retrieval	All dimensions types
15	Web Search Tool	Retrieval	Intuitive
—	Evaluation Logger Tool	System / Logging	System

Table B.1 Consolidated Tool Registry: All 15 MCP Tools with FSLSM Dimension Mapping

Note. The Evaluation Logger Tool is not included in the count of 15 instructional tools as it is a system-level utility and does not participate in tool-selection competition. It operates continuously in the background throughout all experiments.

APPENDIX C
DATASET

C.1 D2L-QA Multi-Hop Synthetic Ground Truth Dataset

Multi-hop question-answer pairs was generated from the Dive into Deep Learning (D2L) textbook (Zhang et al., 2021), where each answer requires synthesizing information across 5 related chunks.

C.1.1 Example Output

{ "question_id": "d2l_mh_0001",

"question": "How do RNNs, attention mechanisms, and scoring functions work together in natural language processing tasks like language modeling and sentiment analysis?",

"gold_answer": "RNNs are foundational in language modeling, as they can predict the next token based on previous tokens, which is exemplified in character-level language models. In tasks like sentiment analysis, bidirectional RNNs enhance representation by considering context from both directions, and pretrained word vectors can further enrich token representation. Attention mechanisms, such as Bahdanau attention, allow the model to selectively focus on relevant parts of the input sequence during decoding, improving the handling of variable-length sequences. Additionally, attention scoring functions like dot product and additive attention are crucial for aggregating information across sequences, enabling more efficient processing in modern architectures like Transformers. The integration of these components allows for sophisticated comparisons between sequences, as seen in natural language inference tasks where aligned tokens are compared using a multi-layer perceptron.",

"answer_type": "synthesize_workflow",

"difficulty": "hard",

"strategy": "intra_cluster",

"quality_tier": "silver",

"gold_chunk_ids": ["309b6eee8af4_0", "c9f7a7f460d0_0", "28a03790ad6c_0", "f56404dbe8d5_0", "0cd70db855b4_0"],

"chunk_contributions": {

"309b6eee8af4_0": "Explains the role of RNNs in predicting tokens in language models.",
"c9f7a7f460d0_0": "Describes how attention mechanisms enhance RNNs by focusing on relevant input parts.",
"28a03790ad6c_0": "Highlights the use of bidirectional RNNs and pretrained vectors in sentiment analysis.",
"f56404dbe8d5_0": "Illustrates the comparison of aligned tokens in natural language inference tasks.",
"0cd70db855b4_0": "Discusses attention scoring functions and their importance in modern architectures." },

"essential_chunk_ids": ["309b6eee8af4_0", "c9f7a7f460d0_0", "28a03790ad6c_0"],

"supporting_chunk_ids": ["f56404dbe8d5_0", "0cd70db855b4_0"]